

WATER EROSION

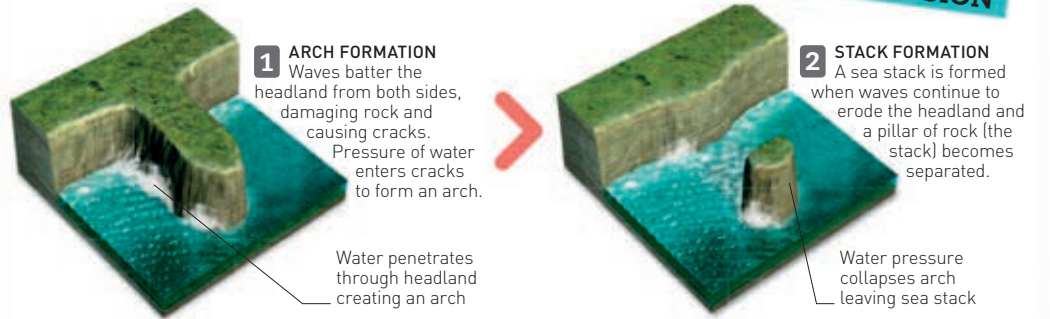
Helped by strong winds, ocean waves batter against coastal landforms. Dislodged rocks and pebbles are ground down and rub abrasively against headlands, cliffs, and standing rocks.



ERODED ROCK, LOCH ARD GORGE, AUSTRALIA

ARCHES AND STACKS

As waves approach a headland, they curve around, attacking the sides. In a process called corrosion, stones flung up by the waves erode the sides, causing cracks. Compressed air brought in by waves expands, enlarging cracks and forming arches and stacks.



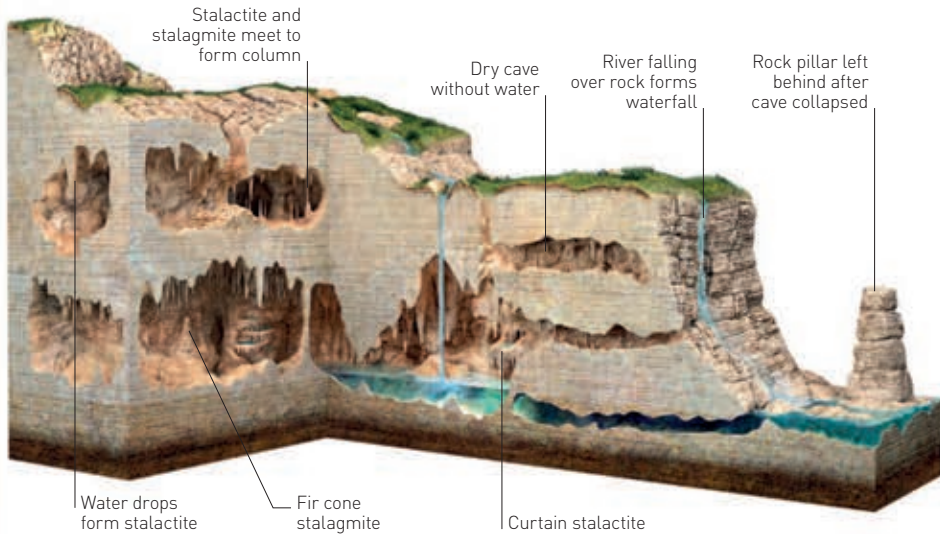
LIMESTONE LANDSCAPES

Deep below Earth's surface are large cave systems. Rainwater, which is slightly acidic, gradually dissolves the limestone, creating cracks. Flowing water widens the cracks, forms channels, and eventually creates cave systems.



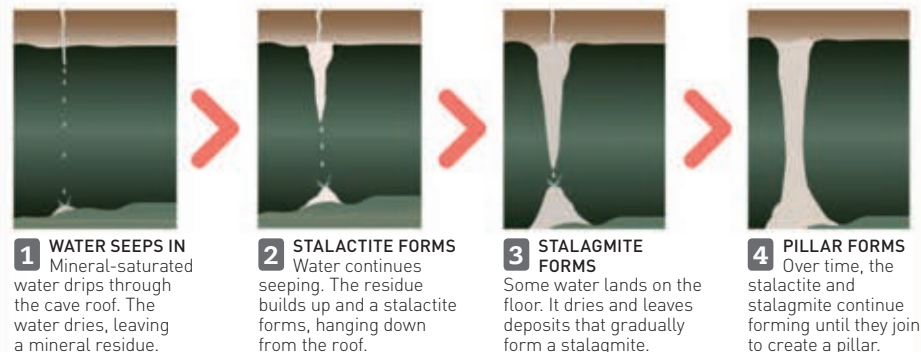
INSIDE A CAVE

Limestone caves are wondrous places. Over centuries, erosion has created huge chambers, often containing many incredibly shaped pillars, and river-filled tunnels. Caves vary in size. Some are shallow but the deepest, in France, lies nearly 1.5 km (1 mile) below ground.



STALACTITES AND STALAGMITES

Slim, beautifully shaped stalactites hang down from the roof of a cave. Stalagmites rise up from the floor of the cave. Both form from a mineral in the rocks called calcite that dissolves in drops of water seeping through the roof. Over time, they form fantastic shapes.



OXBOW LAKES

On low-lying land, snake-like meanders may form when the course of a river bends and may eventually become oxbow lakes. Meanders have two sets of curves: one side is formed by erosion as the river erodes the land, the other side forms from deposits of silt and sediment.



WIND EROSION

Wind is a powerful erosive agent. It blows away soil, sand, and other light substances, depositing them at different locations, often sculpting new landforms. Wind erosion can be destructive, particularly for farmers. Trees and terraces help protect land.



CONICAL FORMATIONS, TURKEY

In some parts of the world the impact of wind has changed landscapes, eroding rocks into new forms.



SANDSTONE SWIRLS, USA

Wind, and the sand particles it carries, erode sedimentary rocks such as sandstone, creating fantastic swirls.

MUSHROOM-SHAPED ROCKS, EGYPT

Desert winds fling sand about, which causes erosion. The sand blasts against rocks, producing strange shapes.

